Technical Report: Energy-Efficient UAV Trajectory Optimization using Particle Swarm Optimization

1. Abstract

This project presents a Python-based simulation for unmanned aerial vehicle (UAV) trajectory optimization using Particle Swarm Optimization (PSO). The objective is to compute a safe and energy-aware path from a start position to a target position while avoiding restricted circular obstacle regions. The work combines mathematical modeling, simulation, optimization, and visualization, making it relevant for UAV research topics such as path planning, flight control, and autonomous systems.

2. Problem Formulation

The UAV trajectory consists of a fixed start point, a fixed goal point, and multiple intermediate waypoints. The optimizer searches for waypoint coordinates that minimize a weighted objective function.

```
J = wd*Distance + we*Energy + ws*Smoothness + wc*CollisionPenalty + wb*BoundaryPenalty
```

The objective includes path length, an energy proxy, a smoothness term, collision penalties for entering obstacle safety zones, and boundary penalties for leaving the mission area.

3. Particle Swarm Optimization Method

PSO is a population-based optimization algorithm inspired by swarm behavior. Each particle represents a candidate UAV trajectory. During each iteration, particles update their positions using their own best known solution and the best solution found by the swarm.

```
v = inertia*v + cognitive*r1*(personal_best - position) + social*r2*(global_best - position)
```

4. Simulation Setup

The mission area is a 10 km x 10 km region with circular obstacle zones representing buildings, towers, or restricted areas. A safety margin is added around each obstacle. The simulation uses 80 particles, 220 iterations, and 7 intermediate waypoints.

5. Results

The optimized trajectory avoids obstacle regions while maintaining a continuous feasible route to the target. The baseline straight-line path is shorter but infeasible because it intersects restricted zones.

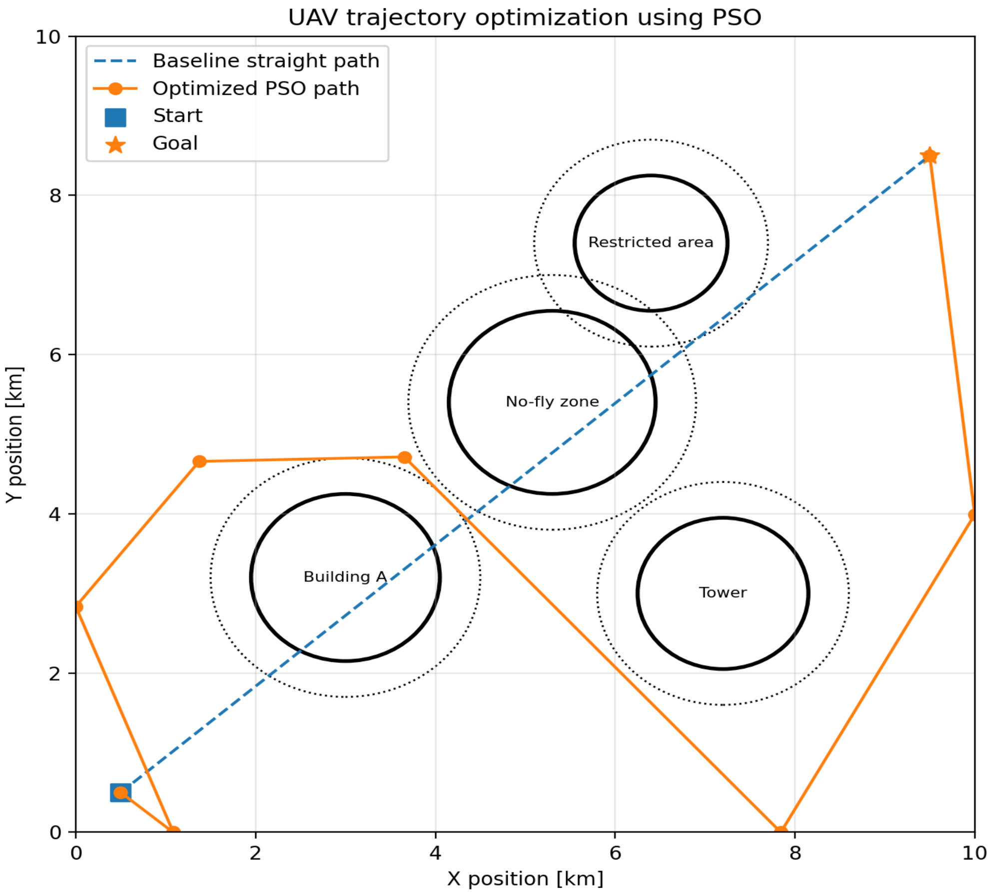

Figure 1: Optimized UAV trajectory compared with baseline straight path.

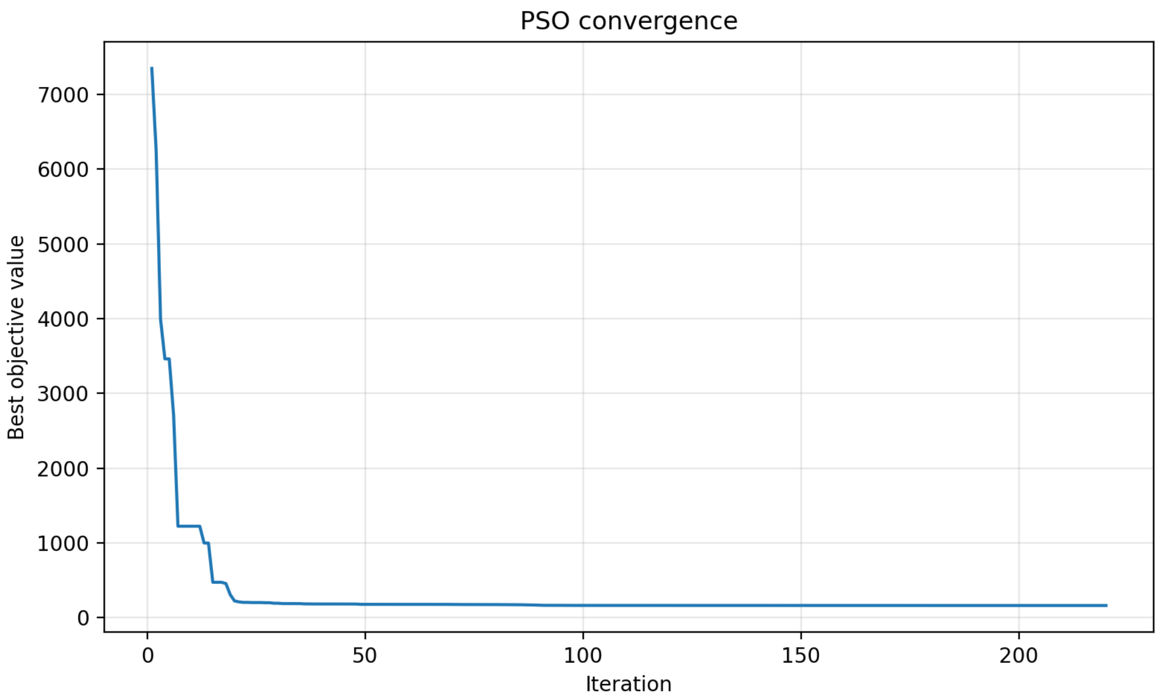

Figure 2: PSO convergence curve.

6. Future Work

Future improvements include 3D trajectory planning, wind disturbance modeling, battery discharge modeling, comparison with Differential Evolution and Genetic Algorithms, PID/LQR trajectory tracking, and integration with ROS 2, PX4, or ArduPilot simulation environments.

7. Conclusion

This project demonstrates a complete workflow for UAV trajectory optimization: mathematical formulation, simulation environment design, PSO implementation, visualization, and result analysis. It is suitable for a GitHub portfolio and for applications involving UAV modeling, simulation, optimization, flight control, and autonomous systems.